Customer Churn Analysis

Project Overview

The objective of this project is to analyse customer behaviour and identify key factors influencing customer churn. Using SQL-based analysis, the project aims to understand patterns in customer data and provide insights to improve retention and business performance.

Problem Statement

Customer churn is a major challenge for businesses as it leads to revenue loss and increased acquisition costs. The problem is to identify why customers are leaving, which segments are at higher risk, and what factors contribute most to churn, so that the company can take proactive actions to reduce it.

Data Source

- **Dataset:** Customer Churn Dataset, Kaggle.com
- **Domain:** Business Analytics
- **Tool Used:** MYSQL

Features

- **Customer_ID:** Unique identifier for each customer
- **Gender:** Gender of the customer
- **Age:** Age of the customer in years
- **Country:** Country where the customer belongs
- **City:** City of the customer
- **Customer_Segment:** Type of customer (Individual, SME, Enterprise)
- **Tenure_Months:** Number of months the customer has stayed with the company
- **Signup_Channel:** Channel through which the customer signed up (Web, Mobile, Referral)
- **Contract_Type:** Type of subscription plan (Monthly, Quarterly, Yearly)
- **Monthly_Logins:** Number of times the customer logs in per month
- **Weekly_Active_Days:** Number of days the customer is active per week
- **Avg_Session_Time:** Average time spent per session
- **Features_Used:** Number of features used by the customer
- **Usage_Growth_Rate:** Growth rate of customer usage over time
- **Last_Login_Days_Ago:** Number of days since last login
- **Monthly_Fee:** Amount paid monthly by the customer
- **Total_Revenue:** Total revenue generated from the customer

- **Payment_Method:** Mode of payment used by the customer
- **Payment_Failures:** Number of failed payment attempts
- **Discount_Applied:** Whether discount was applied (Yes/No)
- **Price_Increase_Last_3M:** Whether price increased in last 3 months (Yes/No)
- **Support_Tickets:** No. of complaints or service requests raised by a customer
- **Avg_Resolution_Time:** Average time taken to resolve issues
- **Complaint_Type:** Type of complaint raised by customer
- **CSAT_Score:** Customer satisfaction score
- **Escalations:** Number of issues passed to higher_level team
- **Email_Open_Rate:** Percentage of emails opened
- **Marketing_Click_Rate:** Percentage of marketing links clicked
- **NPS_Score:** Net Promoter Score indicating customer loyalty
- **Survey_Response:** Customer survey feedback
- **Referral_Count:** Number of referrals made by the customer
- **Churn:** Indicates whether customer churned (1 = Yes, 0 = No)
- **Churned_Status:** Derived column showing churn status (Yes/No)

Queries to do Descriptive Analysis

1. How many total customers are present in the dataset?
Output: 10,000 customers

---

2. How many customers have churned and how many are retained?
Output:

- Retained (No): 8,979
- Churned (Yes): 1,021

---

3. What are the distinct countries represented in the dataset?
Output: Bangladesh, Canada, Germany, Australia, India, USA, UK

---

4. How many customers belong to each signup channel?
Output:

- Web: 5,036
- Mobile: 2,960
- Referral: 2,004

---

5. How many records contain missing CSAT scores?
Output: 0 (No missing values)

---

6. What is the gender distribution of customers?
Output:

- Male: 5,013
- Female: 4,987

---

7. What is the average age of churned vs non-churned customers?
Output:

- Retained (No): 45.85 years
- Churned (Yes): 46.40 years

---

8. What is the total revenue generated by all customers?
Output: 10,570,180

---

9. What is the average monthly fee for each contract type?
Output:

- Monthly: 34.60
- Yearly: 35.47
- Quarterly: 35.11

---

10. What is the average number of monthly logins by churn status?
Output:

- Retained (No): 20.00
- Churned (Yes): 16.80

---

11. Which payment method is most commonly used by customers?
Output: Bank Transfer (1,488 customers)

---

12. What is the average tenure (in months) of customers?
Output: 30.16 months

Queries to do Diagnostic Analysis

13. What is the churn rate for each contract type?
Output:

- Monthly: 4967 customers, 513 churned → 10.33%
- Yearly: 1983 customers, 205 churned → 10.34%
- Quarterly: 3050 customers, 303 churned → 9.93%

---

14. How does average tenure differ between churned and retained customers?
Output:

- Retained (No): 30.83 months
- Churned (Yes): 24.22 months

---

15. What is the relationship between CSAT score and churn?
Output:

- CSAT 1: 23.53% churn
- CSAT 2: 26.02% churn
- CSAT 3: 7.93% churn
- CSAT 4: 6.73% churn
- CSAT 5: 7.72% churn

---

16. Do customers with more support tickets churn more frequently?
Output:

Support_Tickets	Total_Customers	Churned_Customers	Churn_Rate (%)
0	3012	315	10.46
1	3609	353	9.78
2	2124	215	10.12
3	908	108	11.89
4	273	26	9.52
5	56	3	5.36
6	13	0	0.00
7	5	1	20.00

---

17. What is the churn rate by complaint type?
Output:

- Service: 9.95%
- Billing: 10.18%
- Technical: 10.32%
- None: 10.32%

---

18. Does a recent price increase impact customer churn?
Output:

- No increase: 10.17% churn
- Yes increase: 10.39% churn

---

19. Which signup channel has the highest churn rate?
Output:

- Referral: 11.18% churn (highest)

---

20. How does last login activity differ for churned vs retained customers?
Output:

- 0–7 days: 9.95% churn
- 8–30 days: 9.60% churn
- 31+ days: 19.48% churn (highest)

Queries to do Predictive Analysis

21. Potential Churn Risk Customers

Output:
• Risk customers: 1147 (csat_score <= 2)

---

22. Inactive Customers (>30 days)

Output:
• 426 inactive customers identified (last_login_days_ago > 30)

---

23. Does shorter tenure increase churn?

Output:
• Longer Tenure (≥ 6 months): 8.12% churn
• Shorter Tenure (< 6 months): 32.48% churn

---

24. Does low usage increase churn?

Output:
• Higher logins (≥ 5): 8.89% churn
• Lower logins (< 5): 28.59% churn

---

25. High revenue customers who churn

Output:
• 288 High-value churned customers identified (above average revenue group)

Queries to do Prescriptive Analysis

26. Customers to target for retention offers

Output:
• 20 High-risk retained customers identified (top 20 based on risk factors like total_revenue,tenure_months,monthly_logins,csat_score,support_tickets)

---

27. Customers for referral / loyalty programs

Output:
• Top 10 loyal customers identified (high CSAT & referral activity)

---

28. Which segment needs discount pricing?

Output:
• Individual: 9.98% churn
• SME: 10.93% churn
• Enterprise: 9.42% churn

---

29. Which segment needs service improvement?

Output:
• Enterprise: 70.82% support rate
• Individual: 69.95% support rate
• SME: 69.43% support rate

---

30. Which contract type should be promoted?

Output:
• Monthly: 10.33% churn
• Yearly: 10.34% churn
• Quarterly: 9.93% churn

Descriptive Analysis

- Dataset contains 10,000 customers with ~10% churn rate
- Customers are spread across multiple countries with balanced gender distribution
- Majority of customers are acquired through Web channel
- Average tenure is around 30 months and average age is ~46 years
- Retained customers show higher monthly logins and overall revenue is 10.57M

Diagnostic Analysis

- Churn rate is almost similar across contract types (~10%), showing weak impact
- Low tenure and low CSAT strongly increase churn probability
- Low engagement (fewer logins) leads to higher churn
- Inactivity (>30 days) significantly increases churn risk

Predictive Analysis

- 1147 high-risk customers identified for potential churn
- 426 inactive customers flagged as churn-prone
- Short-tenure and low-usage customers are high-risk groups
- 288 high-revenue customers already show churn behaviour

Prescriptive Analysis

- 20 high-risk customers targeted for retention actions
- Loyalty programs recommended for high-CSAT customers
- SME segment requires pricing or engagement improvement
- Monthly contract users need stronger retention strategies

Conclusion

• Customer behaviour (engagement, tenure, inactivity) is the main churn driver
• Early-stage customers and inactive users are highest risk
• Satisfaction strongly impacts retention decisions
• Support ticket analysis shows consistent churn rates across ticket levels, indicating stable support handling, but satisfaction scores are needed to confirm efficiency
• Business should focus on engagement-driven retention strategies